

# Faiz Naufal Huda

Bogor, Indonesia

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## Summary

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Computer Science undergraduate at IPB University (GPA: 3.85/4.00) with hands-on experience building complete, well-tested systems end-to-end, from a full-stack academic document management system (118 tests, Clean Architecture) to an award-winning hackathon e-commerce platform and ML pipelines with CI/CD, Docker, and transformer fine-tuning. Completed the Pijak × IBM SkillsBuild AI Engineer Cohort as a selected participant. Fluent in English (TOEFL ITP 607).

## Technical Skills

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|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>        | Python, JavaScript, TypeScript, SQL                                                                                                            |
| <b>Frontend:</b>         | React 18, Tailwind CSS, Vite                                                                                                                   |
| <b>Backend &amp; DB:</b> | FastAPI, SQLAlchemy 2.0, PostgreSQL, JWT + bcrypt, Alembic                                                                                     |
| <b>ML / Data:</b>        | LightGBM, XGBoost, Scikit-learn, Pandas, NumPy                                                                                                 |
| <b>DevOps / Tools:</b>   | Docker, GitHub Actions (CI/CD), Git, pytest                                                                                                    |
| <b>Familiar:</b>         | Next.js, Supabase, PyTorch, Hugging Face Transformers, TensorFlow/Keras, MLflow, Prometheus, Grafana, Weights & Biases, Bash, ReportLab, PyPDF |

## Projects

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**AQI Bangladesh: Air Quality Prediction** (github.com/faizhuda/daming-kelompok-7) Apr – Jun 2026

- Independently built an end-to-end ML system predicting AQI across 30 cities in Bangladesh; trained LightGBM on 1M+ rows of multi-pollutant sensor data achieving MAE = 1.13, RMSE = 1.77,  $R^2 = 0.9986$ ; chosen over XGBoost for 3–10× faster training on large-scale tabular data.
- Engineered modular pipeline with centralized YAML config, lag features, rolling windows, cyclical time encoding, and walk-forward cross-validation (3-fold) for time-aware evaluation; experiment tracking via CSV logging.
- Implemented 40 unit tests across 6 modules (pytest + coverage); CI/CD via GitHub Actions with black/flake8 enforcement and pre-commit hooks; containerized with Docker for reproducible deployment.

**AgriDesk: Academic Document Management System** (github.com/faizhuda/ads-agridesk) Jan – Jun 2026

- Led full-stack development of a 4-person team project; solely responsible for the complete local implementation: backend in FastAPI + SQLAlchemy 2.0 + Alembic on PostgreSQL and frontend in React 18 + Vite + TailwindCSS v4, built on Clean Architecture, Domain-Driven Design (DDD), and Finite State Machine document workflows with drag-and-drop PDF upload.
- Implemented SHA-256 document hashing with QR code verification enabling third-party authentication without system login; PDF generation (ReportLab + PyPDF) with coordinate-based digital signature stamping.
- Achieved 100% test coverage on core services across 118 passing tests; JWT + bcrypt authentication; in-memory SQLite for isolated unit testing.

**AgritechJam, Agriinformatics 2025 (MyToko)** (github.com/faizhuda/agritechjam-mytoko) Oct – Nov 2025

- Placed 2nd** as full-stack developer in a 3-person team; built a complete customer + admin e-commerce platform using Next.js, TypeScript, Tailwind CSS, and Supabase (PostgreSQL) within hackathon timeframe.
- Implemented PostgreSQL stored procedures for atomic order cancellation with inventory rollback, eliminating race conditions on concurrent stock updates.
- Delivered real-time admin analytics dashboard (Recharts), PDF invoice generation, and full order lifecycle management (Pending → Paid → Shipped → Delivered).

## Experience

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**AI Engineer Cohort (Selected Participant)**

Jan – Jul 2026

Pijak × IBM SkillsBuild (Dicoding Indonesia)

- Completed structured learning path covering Python, supervised/unsupervised ML, deep learning (CNN & RNN), experiment tracking (MLflow), and production ML deployment via IBM SkillsBuild and Dicoding's industry curriculum.
- Built **PriorMail** as capstone: an email intelligence system with DistilBERT fine-tuning for 4-class priority classification; designed phishing detection module with security-focused adversarial evaluation protocol.
- Earned dual certificates from IBM SkillsBuild and Dicoding upon module completion.

**Head of Funding & Consumption Division**

Apr 2025 – Dec 2025

IT TODAY, HIMALKOM IPB University

- Managed IDR 33 million F&B consumption budget covering meals, snacks, and catering for 500+ participants and 90+ committee members across the full IT TODAY event series.
- Secured IDR 5 million in internal operational funding through entrepreneurial methods to support division activities.

**Treasurer, Welfare & Internal Harmony Division**

Jan 2025 – Dec 2025

HIMALKOM IPB University

- Managed annual divisional budget of ~IDR 15–20 million across 2–3 major programs, achieving 100% budget absorption with zero deficit, reflecting accurate planning and real-time expenditure tracking.
- Oversaw 20–30 financial transactions per program cycle; handled reimbursements and documentation with full accountability and transparent reporting.

**Teaching Volunteer**

Feb 2025 – Nov 2025

LPK Tepi Sawah, HIMALKOM IPB University

- Delivered 14 teaching sessions to 25+ children in Leuwiliang, Bogor over 9 months, covering literacy, English, mathematics, environmental awareness, and introductory programming through game-based activities.
- Served a dual role: committee organizer (coordination, logistics) and on-site volunteer (direct teaching and mentoring).

**Education**

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**IPB University**

Jul 2023 – Oct 2027 (Expected)

Bachelor of Computer Science

GPA: 3.85 / 4.00

**Certifications**

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- **AI Engineer Cohort**, IBM SkillsBuild + Dicoding Indonesia Jan – Jul 2026
- **Associate Data Scientist**, BNSP (Badan Nasional Sertifikasi Profesi) Jul 2026
- **Belajar Fundamental Deep Learning** (Deep Learning Fundamentals), Dicoding Indonesia May 2026
- **Belajar Machine Learning untuk Pemula** (ML for Beginners), Dicoding Indonesia Mar 2026
- **Python for Data Science and ML Essential Training (Part 1 & 2)**, LinkedIn Learning Feb 2026
- **TOEFL ITP**, ETS (Score: 607) Nov 2023